PM2.5, PM10 ~ station + time, analysis

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It is common to find panel data in real world setting where it can sometimes be very noisy and have gaps. Such a situation can be encountered when working with local price indices, sensor data and small area statistics. Oftentimes a smooth underlying level is desired for each series including periods where nothing was observed. This work builds a small PyTorch network and models two related targets jointly treating the missing of the two as masked instead of dropping the row. It weights each observation by number of readings used to deduce the average and ensures smoothness with a curvature penalty. To evaluate the approach, hourly PM2.5 and PM10 dataset from 12 Beijing monitoring stations is used. It is thinned systematically to introduce sparsity and the model is scored against full oberved data. The neural net (using a joint model) fills long gaps and very sparse station data better than if smoothing was applied to series independently. Jointly modelling PM2.5 and PM10 also recovers a completely dropped series, but can lose some accuracy compared to individual series model in data rich instances.

Table of contents

1 Introduction 2

2 Data 2

3 Model 3
- 3.1 Architecture . . . 3
- 3.2 Loss . . . 3

4 Experiment 4
- 4.1 Choosing the settings . . . 4
- 4.2 Results against the 31 day MA . . . 5
- 4.3 What the fits look like . . . 5
- 4.4 A bias to fix . . . 8

5 Takeaways 8

> **Note**
>
> This work was developed with AI assistance

1 Introduction

To begin with we'll define a panel dataset where in each location has its own timeseries of two quantities of interest which are related (e.g. could be 2 sensors in vicinity or a location with rent and sale prices etc.). Each observation is a mean taken over raw readings which makes the data noisy. Some locations may have data on only one of the series while others may not have either. Some locations may be missing contiguous data points over several time periods. We are interested in a *prevailing level* which is a smooth estimate of both series at each location which is robust to noise and (partial/full) missing data.

An approach that focuses on modelling series' independently falls short when there are long gaps in series or the series is overwhelmingly noisy. Ironically, these are the situations where a good estimate is expected of a method.

It is obvious that some kind of information sharing is required for robustness and there are 2 that are available in the kind of panel chosen for this study:

- **across locations** nearby or similar locations are more related, so a sparse/noisy series can borrow from its peer locations
- **across targets** the two quantities are related, so a month where only one was observed can inform about the other

The approach below does both in one network, and the experiment measures how each is influential in ensuring robustness of estimate.

2 Data

To be able to work with sparse data but still have some ground truth for evaluating our approach, we start from the full UCI *Beijing Multi-Site Air Quality* dataset which is an hourly PM2.5 and PM10 measurement data at 12 stations spanning March 2013 - February 2017. Both the pollutants are closely related and are a good fit for the panel data setting we hope to address with our work.

The analysis focuses on daily data at each station by taking the mean value of hourly readings in the day. Number of hours where data is available in the day n is retained as a measure of reliability of mean. The full dataset is very rich (>99% station days coverage) so we thin it to make it noisy and introduce missingness using the following procedure:

- For each station a random Posson($\lambda = 1.2$) number of hours are kept (draws separately for PM2.5 and PM10). This can often lead to situation where only either is observed on any given hour
- From each station and pollutant series we remove two blocks of 20–60 (random) consecutive days

The **ground truth** is the full data which is then used in evaluation, by comparing estimates against a 31 day moving average of daily (log) means of ground truth.

pollutant	full: days observed	full: hours per day	sparse: days observed	sparse: hours per day
PM2.5	100%	23.6	66%	1.7
PM10	100%	23.7	67%	1.7

In the thinned data both PM2.5 and PM10 are observed on 44% station days and exactly one of them is observed on 45%. The figure shows one station over a year as a result of thinning the data.

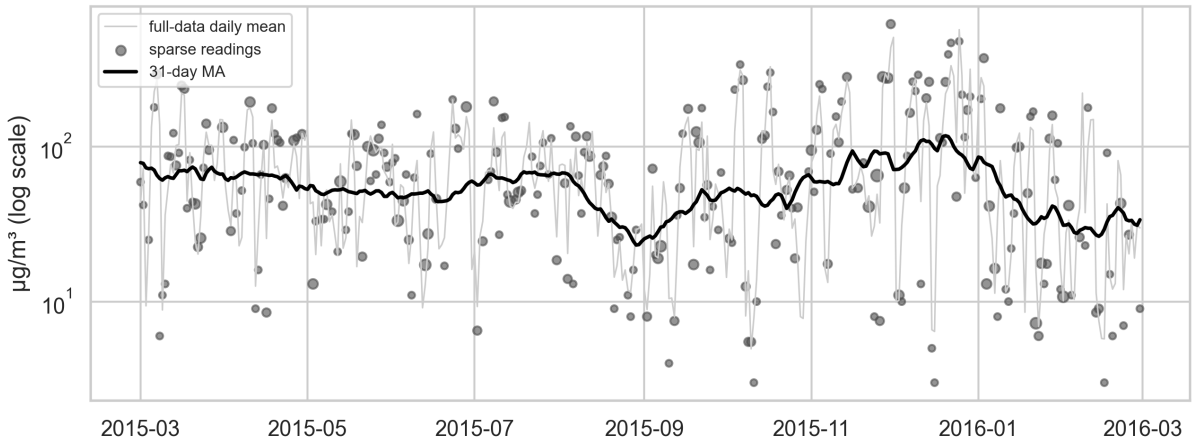


Figure 1: Dongsi station, PM2.5, one year. Grey: full-data daily means. Dots: the sparse readings (size = hours). Black: the 31-day MA of the full data.

Daily pollution varies by a factor of two or more from day to day with the weather and a reading of only one or two hours adds large degree of noise. The *prevailing level* is the slow black line.

3 Model

3.1 Architecture

Every row is one station day with up to two targets: $y_1 = \log \mathrm{PM2.5}$ and $y_2 = \log \mathrm{PM10}$ (standardised). The network maps *(station, day)* to a predicted level for both:

$$
\begin{aligned}
x &= \left[\, e_{\text{area}} + e_{\text{station}},\ b(t) \,\right] && \text{8-dim location embedding, concatenated with the time bumps} \\
h &= \mathrm{SiLU}(W_2\, \mathrm{SiLU}(W_1 x)) && \text{shared trunk, two layers of 64} \\
\mu_1 &= w_1^\top h, \quad \mu_2 = w_2^\top h && \text{one linear head each for PM2.5 and PM10}
\end{aligned}
$$

- **Location** is the area's embedding plus a station offset that starts at zero and is penalised, so a station with less data falls back to its area
- **Time** enters only through fixed Gaussian bumps, $b_k(t) = \exp\{-\frac{1}{2}((t - c_k)/w)^2\}$ with centres c_k every w days. A day becomes a vector of bump values; neighbouring days get nearly identical vectors, so the level cannot jump from one day to the next.
- **A shared trunk** of two layers with a smooth activation (SiLU) turns location and time into a 64 digit summary; **two linear heads** read the same summary. Everything before the heads is shared, so a day that observed only PM2.5 still trains everything PM10 depends on.

3.2 Loss

Each daily observation is a mean of n hours worth of readings, consequently the variance is readling variance divided by n, plus a floor τ^2 for daily movements the *prevailing level* should not follow. For target j:

$$\mathcal{L}_{\text{fit}} = \frac{1}{\sum m_{ij}} \sum_{i,j} m_{ij} \left[\tfrac{1}{2} \log v_{ij} + \frac{(y_{ij} - \mu_{ij})^2}{2 v_{ij}} \right], \qquad v_{ij} = \frac{\sigma_j^2}{n_{ij}} + \tau_j^2,$$

where m_{ij} is 1 if target j was observed on row i and 0 otherwise — a missing target is **masked**, not dropped. σ_j and τ_j are learned, one per target. A day of 24 readings counts far more than a day of one.

Smoothness comes from a penalty on the curvature of every station's predicted series over the **whole** day grid, observed or not:

$$\mathcal{L} = \mathcal{L}_{\text{fit}} + \lambda \overline{(\mu_{t+1} - 2\mu_t + \mu_{t-1})^2} + \gamma \overline{\|e_{\text{station}}\|^2}.$$

The penalty covers days without data (since predicted values exist for those periods), it controls the curve inside gaps and at the edges.

4 Experiment

To evaluate model on worst case of data corruption, two stations are degraded further before fitting:

- **a dropped series** every PM10 reading at *Dingling* is removed retaining only its PM2.5
- **a very sparse station** *Huairou* retains only 15% of its (already sparse) days

Three estimators are fitted to the same data:

1. **joint** the network we're building, modelling targets jointly
2. **single-target** the same network trained on one pollutant at a time (two separate models)
3. **±15-day mean** for each station and pollutant, the hour-weighted mean of the sparse readings within 15 days. Where a station has no readings at all, the same within its area.

4.1 Choosing the settings

The penalty weight λ, bump spacing w and number of training steps are chosen by look ahead cross validation (`cv.py`). There are 12 sequential folds, 30 days apart over the final year. Each trains on everything up to its cutoff and is scored on the next 15 days of sparse readings, with **every station hidden at once**. The chosen settings are $\lambda = 100000$, $w = 30$ days and 100 training steps.

We also tried hiding one station per area while its neighbours kept reporting. The two schemes ask different questions:

- **Every station hidden:** the best guess for the next 15 days is whatever persists, the slow seasonal part of the series. Episodes of two to three weeks don't carry past the cutoff, so a curve that follows them is punished. This defines the level as *the part of the series that persists into the near future.*
- **One station hidden:** asks what this station is doing right now, given its neighbours right now. That rewards following shared short episodes, and picks a much rougher curve.

Neither is wrong, but a *prevailing level* is the first, the overall level, not the episodes.

4.2 Results against the 31 day MA

Every station day and pollutant is assigned to one group:

- **dropped series** Dingling PM10 (no data at all)
- **very sparse station** Huairou, both pollutants present
- for every other station: **observed** (the sparse data has readings that day), **unobserved** (no readings, exlcuding the gap block), **long gap** (inside a 20–60 day gap block) and among unobserved days those where **the other pollutant was observed**.

model group	station-days	joint	single-target	±15-day mean
observed	20320	0.154	0.147	0.153
unobserved	8608	0.153	0.146	0.158
other pollutant observed	6544	0.152	0.145	0.178
long gap	1753	0.156	0.144	0.292
very sparse station	2922	0.152	0.141	0.496
dropped series	1461	0.144	0.182	0.156

We measure Mean Absolute Error (on log data) from the 31 day MA (0.15 15%). When the day is observed, the raw sparse reading itself has approx. MAE 0.7-0.8, so the three estimators remove most of the noise. They behave differently in hard cases as follows:

- **Long gaps:** the networks carry the level through (joint 0.156, single target 0.144), while ±15 day MA runs out of data (0.292).
- **Very sparse station:** with only 15% of its days, Huairou's data is too thin for a moving average (0.496). Both networks do as well as on regular stations (joint 0.152, single target 0.141), because the strong smoothing and the shared area pattern fill in what the station lacks.
- **Dropped series:** this is where modelling both pollutants together pays off. With no PM10 at Dingling at all, the joint model infers it from Dingling's PM2.5 (0.144), while a single target model can only fall back to the area (0.182). The joint model also beats the area moving average (0.156).
- **Everywhere else, sharing costs a little.** On observed, unobserved and long gap days, the single target models are slightly more accurate than the joint model (for example 0.147 vs 0.154 on observed days). When a station has its own data for a pollutant, borrowing from the other pollutant adds little and forces one shared trunk to serve both.
- **Densely observed days:** the simple ±15 day MA is comparable to the networks (0.153). Its 31 day window is the same as the benchmark's, which flatters it.

Pollution at all stations also moves together from day to day (weather is citywide). The moving averages pick that up from same day readings, while the network's only notion of time is a smooth curve. A shared day level effect in the network is the obvious next step.

4.3 What the fits look like

A well observed station. Over a year at Dongsi, all three estimators follow the level. With this much data the moving average tracks it closely. The network curves are much smoother and they follow the seasons but flatten shorter episodes, such as the December 2015 peak.

A very sparse station. With only 15% of its days left, Huairou's data is too thin for the moving average, which jumps around. Both networks lean on the suburban area and the shared time pattern and stay close to the level, the joint and single target curves are very similar.

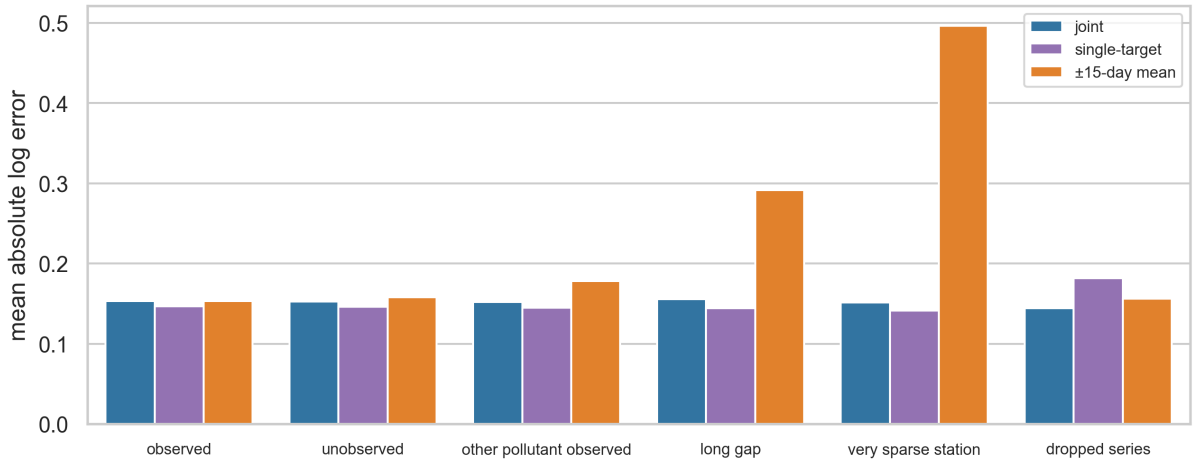


Figure 2: Error against the 31-day MA by group and estimator.

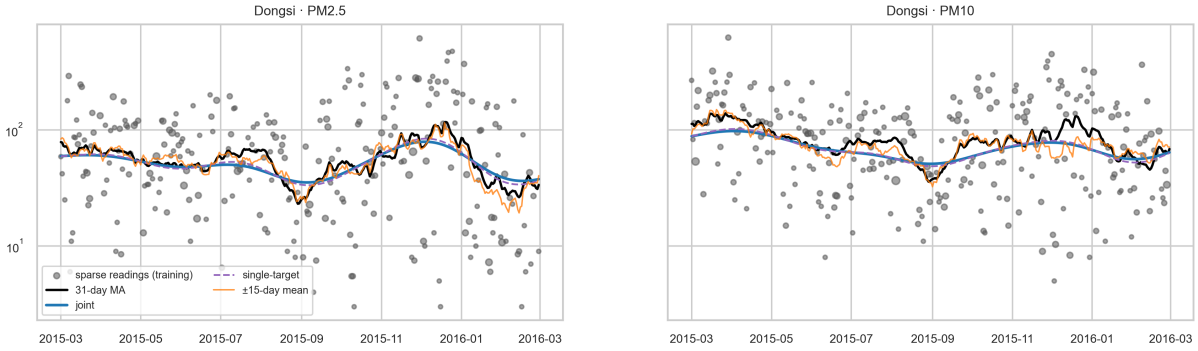


Figure 3: Dongsi, both pollutants, one year (regular sparse data).

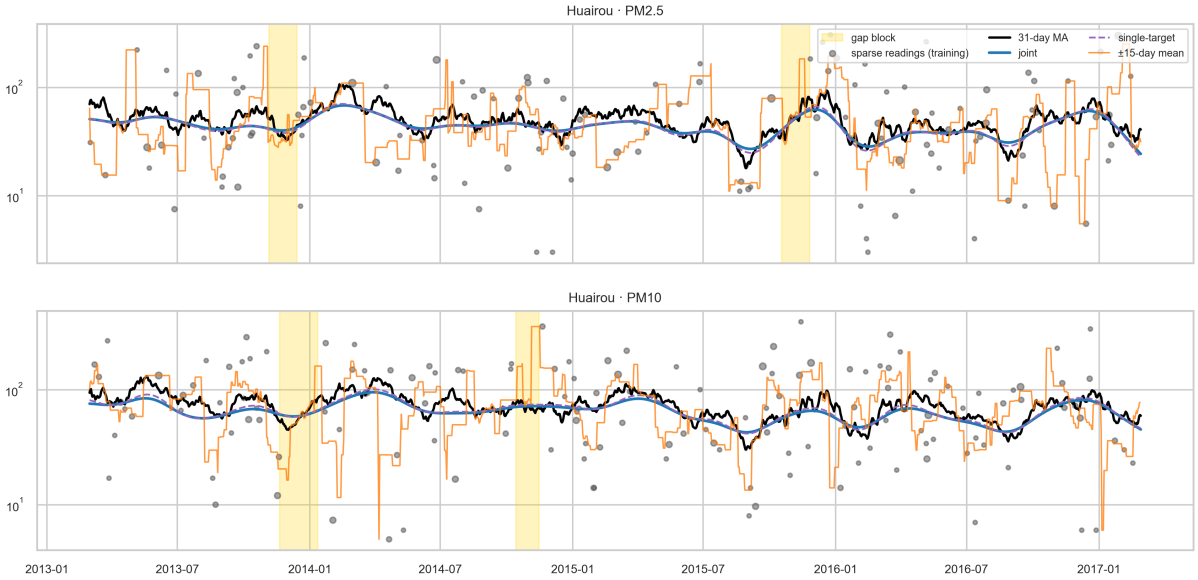


Figure 4: Huairou with 15% of its sparse days kept.

A dropped series. Dingling's PM10 was never observed. The joint model reconstructs it from Dingling's PM2.5 (top). The single target PM10 model has nothing station specific to continue from and returns the suburban area's level, which is too high. The joint curve follows the shape but runs a little low and misses some peaks. The orange line here is the area moving average, which tracks daily movements from the other suburban stations.

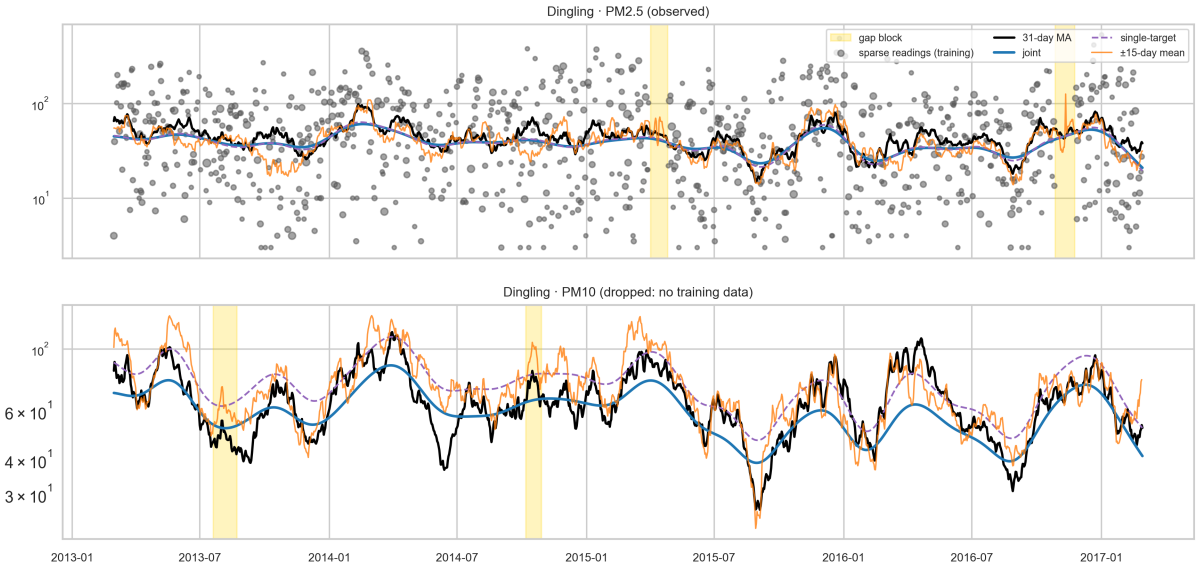


Figure 5: Dingling: PM2.5 observed (top), PM10 completely dropped (bottom).

Long gaps. The four longest gap blocks at regular stations. Once its window empties, the moving average either stops or swings on the few readings at its edges; the networks carry on smoothly. The joint and single target curves overlap, and both sit slightly below the level (see the bias below).

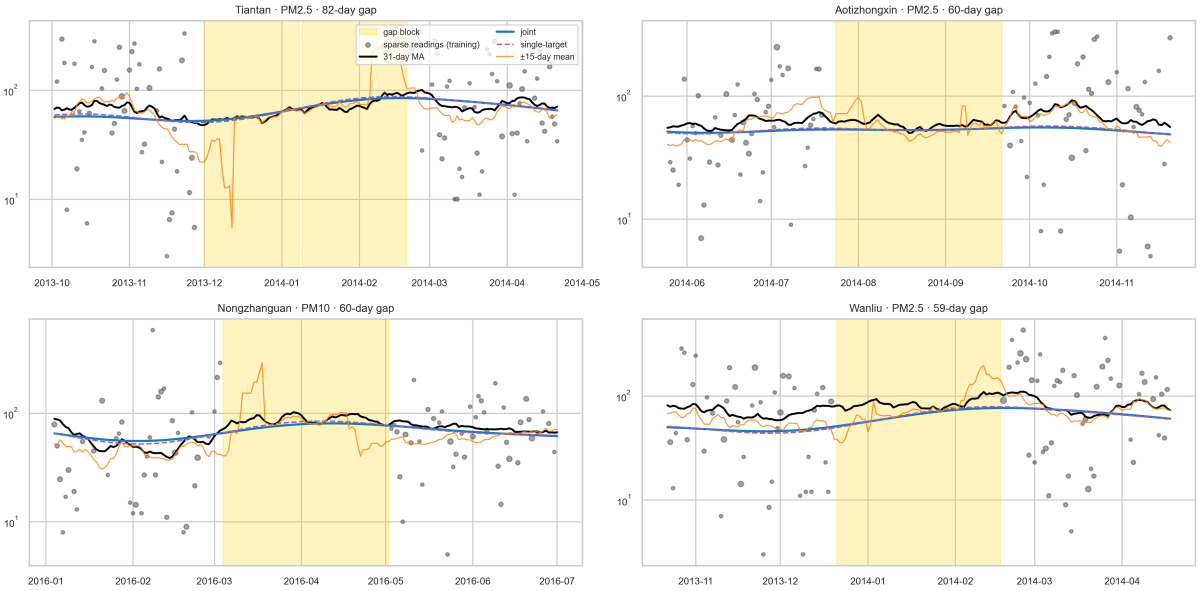


Figure 6: The four longest gap blocks (shaded) at regular stations, with 60 days either side.

4.4 A bias to fix

The model outputs are consistently lower than 31 day MA, by 0.11 on log scale on average (about 11%). The model is fitted to log of means over 1-2 hours and the log of a mean over few readings is lower on average compared to the log of a mean over 24 hours (Jensen's inequality). The typical lognormal correction ($+\frac{1}{2}\sigma^2$) overshoots by about as much in the other direction since the reading level spread mixes within day and between day variation.

5 Takeaways

- **Mask, don't drop.** Treating a missing target as missing in the loss, rather than discarding the row, is what lets one target fill in for the other. It paid off for an entirely missing series.
- **Weight by information.** Variance $\sigma^2/n + \tau^2$ lets a 24-hour mean outweigh a single reading, and stops the model from chasing unnecessary daily movements.
- **Make smoothness explicit.** Smooth time bumps plus a curvature penalty over every period, observed or unobserved, keep the level stable through gaps and at the edges.
- **Share across locations with offsets.** A thin station falls back to its area instead of doing nothing.
- **Know what the network cannot see.** With time entering only as a smooth curve, it misses shared daily shocks that a simple same day average catches. If data is dense, simple smoothing is hard to beat.
- **Bumps alone don't make a curve smooth where there is no data.** Without the curvature penalty, filling gaps inside the data improves slightly (0.13 vs 0.15 on the test gaps). But looking 15 days past the end, the error jumps from 0.25 to 0.5–2.9. The terminal bumps have no data to pin them down. Training length then becomes the only brake, and running 1,000 steps instead of 100 nearly doubles the error.
- **Decide the smoothness, don't tune it.** Validated on noisy readings, a weaker penalty always looks slightly better because it follows real short episodes. How smooth the level should be is a choice about what the level means. Freeze it, and let cross validation pick the rest.
- **A mixed effects GAM does as well or better, when its smoothness is set manually.** Area curves with station deviations shrunk towards them, fitted jointly to both pollutants, beat the network in every group when the smoothness is set by hand. Left to REML it follows weekly swings and does much worse. The extra error is variance, not bias.
- **The tuning target, not the model, sets the smoothness.** Tuned with the same lookahead CV as the network, the GAM is pushed to a nearly flat curve and does worse than the network. The CV only checks the 15 days just past the end of the data, where any flexible spline goes wrong, so it picks the stiffest one. I[]'t drives both models towards oversmoothing.

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